

Hostel Accommodation and Student Services Management System

Team Members

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Course Details

Section: CSE Rhinos

Academic Year: 2026–2027

Team Size: 2 Members

Problem Statement, Objectives & Scope

Problem Statement

Manual or disconnected hostel records for students, rooms, occupancy, payments, complaints and services cause data duplication, hard-to-track room availability, allocation errors and inefficient record management.

Intended Users

Hostel Administrators • Wardens • University Administration

Objectives

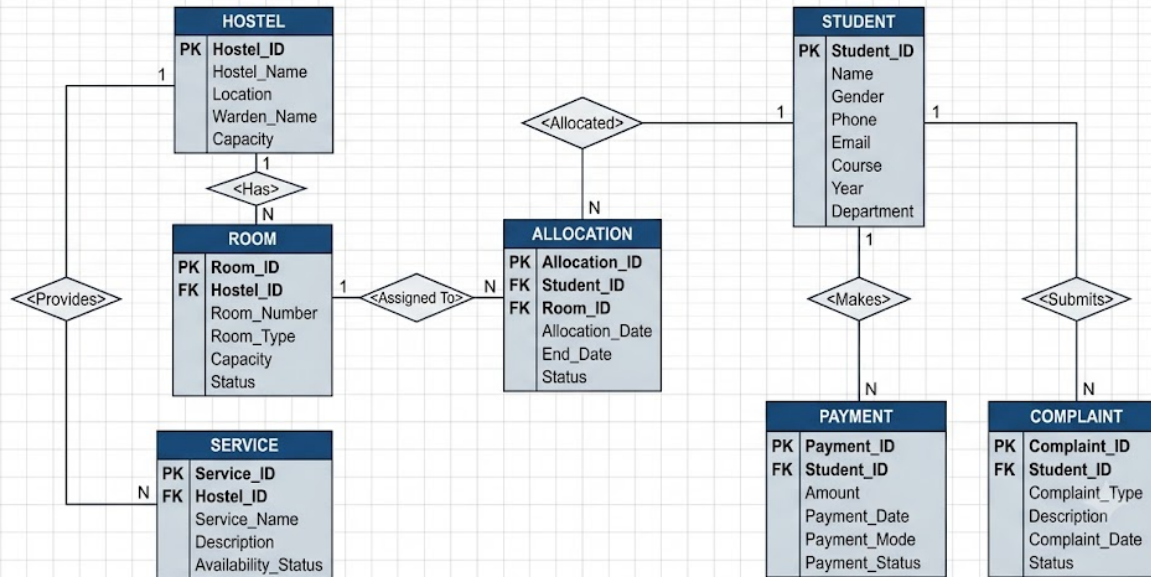
- Centralize hostel and student information
- Manage rooms and allocations efficiently
- Track student payments
- Manage complaints and hostel services
- Provide CRUD operations through a UI
- Maintain integrity using primary & foreign keys
- Reduce manual record management

Scope



ER Diagram & Database Relationships

ER Diagram — Hostel Accommodation and Student Services Management System



PK = Primary Key FK = Foreign Key

7 Entities

Student • Hostel • Room • Allocation •
Payment • Complaint • Service

Relationships

Hostel → Room 1:N

Student → Allocation 1:N

Room → Allocation 1:N

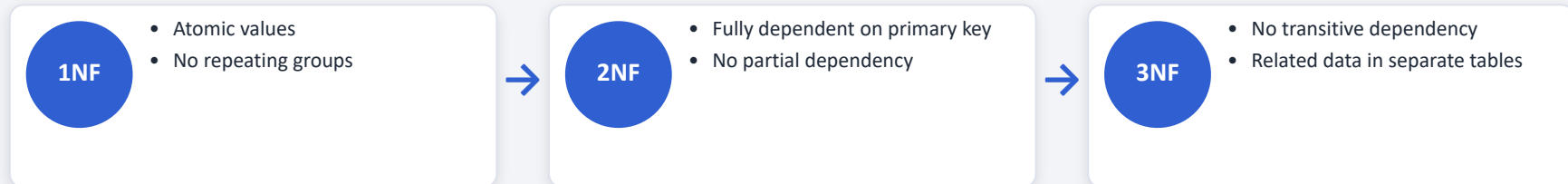
Student → Payment 1:N

Student → Complaint 1:N

Hostel → Service 1:N

Relational Schema & Normalization

STUDENT	HOSTEL	ROOM	ALLOCATION
Student_ID (PK) , Name, Gender, Phone, Email, Course, Year, Department	Hostel_ID (PK) , Hostel_Name, Location, Warden_Name, Capacity	Room_ID (PK) , Hostel_ID (FK) , Room_Number, Room_Type, Capacity, Status	Allocation_ID (PK) , Student_ID (FK) , Room_ID (FK) , Allocation_Date, End_Date, Status
PAYMENT	COMPLAINT	SERVICE	Legend
Payment_ID (PK) , Student_ID (FK) , Amount, Payment_Date, Payment_Mode, Payment_Status	Complaint_ID (PK) , Student_ID (FK) , Complaint_Type, Description, Complaint_Date, Status	Service_ID (PK) , Hostel_ID (FK) , Service_Name, Description, Availability_Status	PK = Primary Key FK = Foreign Key



Database normalized up to 3NF to reduce redundancy and maintain data integrity.

Database Implementation

SCHEMAS

▼ hostelmanagement

▼ Tables

- student
- hostel
- room
- allocation
- payment
- complaint
- service

```
CREATE DATABASE hostelmanagement;  
USE hostelmanagement;
```

```
CREATE TABLE Student (  
  Student_ID INT PRIMARY KEY,  
  Name VARCHAR(100) NOT NULL,  
  Gender VARCHAR(10) NOT NULL,  
  Phone VARCHAR(15) UNIQUE,  
  Email VARCHAR(100) UNIQUE,  
  Course VARCHAR(50) NOT NULL,  
  Year INT NOT NULL,  
  Department VARCHAR(50) NOT NULL  
);
```

✓ Student table created successfully

✓ 6 more tables created: hostel, room, allocation, payment, complaint, service

Constraints Used

PRIMARY KEY

FOREIGN KEY

NOT NULL

UNIQUE

7

Tables

5+

Records per Table

35+

Total Records

SQL Queries, Joins & Aggregate Functions

JOIN QUERY

```
SELECT s.Name, r.Room_Number, h.Hostel_Name
FROM Student s
JOIN Allocation a
    ON s.Student_ID = a.Student_ID
JOIN Room r
    ON a.Room_ID = r.Room_ID
JOIN Hostel h
    ON r.Hostel_ID = h.Hostel_ID;
```

Name	Room Number	Hostel
Rahul Sharma	A-101	Krishna Hostel
Arjun Reddy	A-102	Krishna Hostel
Priya Sharma	B-201	Godavari Hostel
Kiran Kumar	B-202	Godavari Hostel
Sneha Reddy	C-301	Krishna Hostel

AGGREGATE QUERY

```
SELECT
    COUNT(*) AS Total_Students,
    SUM(Amount) AS Total_Payment,
    AVG(Amount) AS Average_Payment
FROM Payment;
```

RESULT

Total Students	Total Payment	Average Payment
5	₹62,500	₹12,500

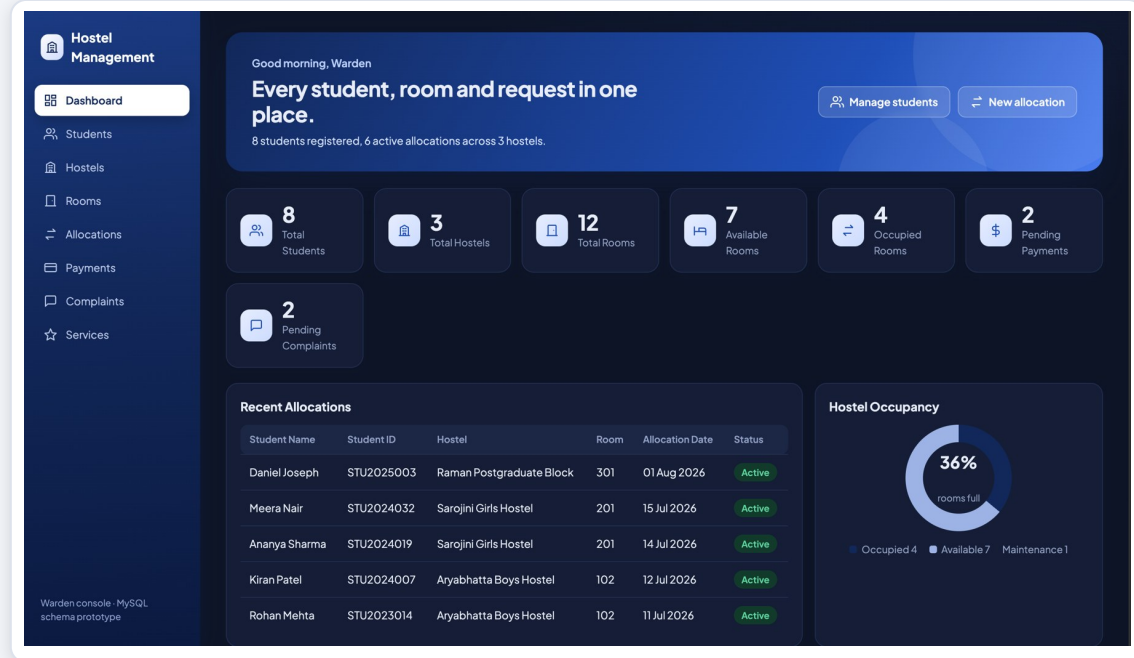
COUNT() | SUM() | AVG() | GROUP BY

GROUP BY EXAMPLE

```
SELECT Payment_Status, COUNT(*)
FROM Payment
GROUP BY Payment_Status;
```

Note: these are sample outputs for presentation demonstration.

User Interface & CRUD Operations



Modules

Dashboard • Student Management • Hostel Management • Room Management • Allocation Management • Payment Management • Complaint Management • Hostel Services

CRUD Operations

INSERT

Add Student → Submit → Record Added

VIEW

Student Management → View → Student Details

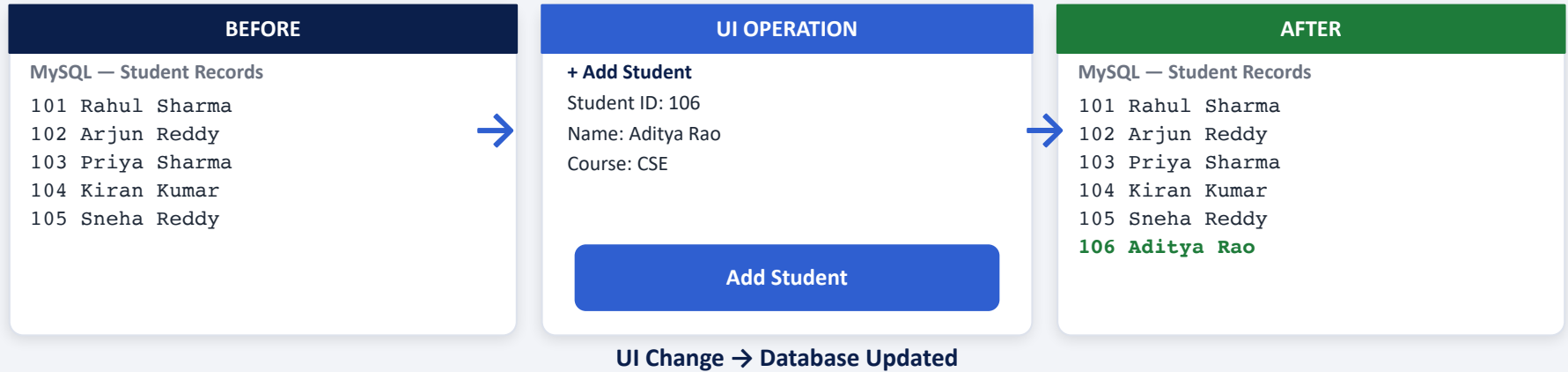
DELETE

Select Student → Delete → Record Removed



The UI is connected to the MySQL database and supports insertion, deletion and viewing of records.

Database Synchronization & Conclusion



Conclusion

- Centralized hostel management; reduced data redundancy
- Improved room and allocation management
- Secure relationships using PK/FK
- Normalized up to 3NF
- Supports CRUD, JOIN and aggregate queries

Future Enhancements

- Online payment integration
- Student authentication
- Email/SMS notifications
- Mobile application
- Advanced analytics

Thank You